D. Manhattan Circle

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 900 | Tags: implementation, math | https://codeforces.com/problemset/problem/1985/D

Given a n by m grid consisting of '.' and '#' characters, there exists a whole manhattan circle on the grid. The top left corner of the grid has coordinates (1,1), and the bottom right corner has coordinates (n, m).

Point (a, b) belongs to the manhattan circle centered at (h, k) if $|h - a| + |k - b|$ lt r, where r is a positive constant.

On the grid, the set of points that are part of the manhattan circle is marked as '#'. Find the coordinates of the center of the circle.

Input

The first line contains t ($1 \le t \le 1000$) — the number of test cases.

The first line of each test case contains n and m ($1 \le n \cdot m \le 2 \cdot 10^5$) — the height and width of the grid, respectively.

The next n lines contains m characters '.' or '#'. If the character is '#', then the point is part of the manhattan circle.

It is guaranteed the sum of $n \cdot m$ over all test cases does not exceed $2 \cdot 10^5$, and there is a whole manhattan circle on the grid.

Output

For each test case, output the two integers, the coordinates of the center of the circle.

Example

Input

```
6
5 5
.....
.....
..#..
.....
.....
5 5
..#..
.###.
#####
.###.
..#..
5 6
......
......
.#....
###...
.#....
1 1
#
5 6
...#..
..###.
.#####
..###.
...#..
2 10
..........
...#......
```

Output

```
3 3
3 3
4 2
1 1
3 4
2 4
```